claimed invention (see also T 2297/10, T 1199/16, T 1341/16). An effect could not be validly used in the formulation of the technical problem if the effect required additional information not at the disposal of the skilled person even after taking into account the content of the application in question. See also T 584/10.

In T 377/14 the board, referring to T 344/89, held that the problem did not have to be explicitly disclosed in the application as filed; it sufficed if it was foreshadowed therein (see also T 478/17).

In T 1841/11 the board held that a problem which is not mentioned in the application in relation to the claimed feature, and which would not arise over the whole ambit of the claim, or even for those embodiments described in detail in the application, cannot be considered to be a suitable choice.

In T 943/13 the board came to the conclusion that the causal relationship between the substance or composition on the one hand and the therapeutic effect achieved on the other hand was decisive for the assessment of inventive step of further-medical-use claims. This causal relationship constitutes the claim's contribution over the prior art. Accordingly, the inventive step of such a claim hinges on the question as to whether this causal relationship, and not just the substance or composition as defined in the claim, is obvious. Hence, in the case in hand the board considered that the objective technical problem was the provision of the claimed therapeutic effect by a different means.

4.2. Formulation of the objective technical problem

The objective technical problem determines the angle of vision that the skilled person will adopt when considering the remaining prior art in the third step of the problem-and-solution approach. For a fair and objective assessment of inventive step, it is therefore important that the objective technical problem is formulated neither too narrowly nor too broadly. In most cases, the objective technical problem can be formulated as how to achieve the technical effect. It is usually a problem the skilled person is familiar with because it relates to known drawbacks of the prior art in the technical field of the invention (T 1448/15).

The technical problem may be formulated using an aim which is to be achieved in a **non-technical field**, and which is thus not part of the technical contribution provided by the invention to the prior art (see e.g. G 1/19, OJ 2021, A77; T 641/00, OJ 2003, 352; T 154/04, OJ 2008, 46). See also in this chapter I.D.9.1. "Assessment of inventive step in the case of mixed-type inventions".

In identifying the problem it is not permissible to draw on knowledge acquired only after the date of filing or priority. According to T 268/89 (OJ 1994, 50) the non-effectiveness of a prior art apparatus or method recognised or alleged only after the priority or filing date could not be drawn on in **formulating the problem**, particularly where that problem was adduced in support of inventive step in a "problem invention" (cf. T 2/83, OJ 1984, 265).

It must be examined whether the problem defined by reference to the closest prior art has indeed been solved the claimed invention. If not, the problem must be reformulated. See also in this chapter I.D.4.4 "Reformulation of the technical problem".

See also Guidelines G-VII, 5.2 – March 2022 version.

4.2.1 No pointer to the solution

According to the established case law, the technical problem addressed by an invention has to be formulated in such a way that it does not contain pointers to the solution or partially anticipate the solution, since including part of a solution offered by an invention in the statement of the problem necessarily results in an ex post facto view being taken of inventive step when the state of the art was assessed in terms of that problem (see e.g. **T 229/85**, OJ 1987, 237; **T 99/85**, OJ 1987, 413; **T 289/91**, OJ 1994, 649; **T 986/96**; **T 799/02**; **T 2461/11**; **T 1252/14**; **T 1230/15**; **T 2690/16**, **T 686/18**).

In **T 800/91** the board emphasised that the formulated problem should be one which the skilled person knowing only the prior art would wish to solve. It should not be tendentiously formulated in a way that unfairly directed development towards the claimed solution.

In **T 1019/99** the board stated that the correct procedure for formulating the problem was to choose a problem based on the technical effect of exactly those features distinguishing the claim from the prior art that was as specific as possible without containing elements or pointers to the solution (cited as established case law in e.g., **T 698/10**, **T 826/10**, **T 143/12**; see also **T 1557/07**, **T 97/13**, **T 1230/15**, **T 67/16**, **T 1861/16**).

In **T 910/90** the board stated that the technical task of an invention must be formulated in such a way that it does not contain any solution ideas; when assessing the objective problem, the closest prior art and any technical progress achieved by the characterising features of the invention had to be taken into account. In so doing, it was not important whether the objective problem had already been mentioned in the closest prior art; what mattered was what the skilled person objectively recognised as the problem when comparing the closest prior art with the invention. See also **T 214/01**.

4.2.2 Problem formulated in the patent application as starting point

According to the established case law, an objective definition of the problem to be solved by the invention should normally start from the problem described in the application/contested patent. Only if examination shows that the problem disclosed was not solved or if inappropriate prior art was used to define the problem, is it necessary to investigate which other problem objectively exists (see e.g. **T 495/91**, **T 881/92**, **T 419/93**, **T 606/99**, **T 728/01**, **T 1708/06**, **T 1146/07**, **T 1060/11**, **T 204/16**). In **T 400/98**, the technical problem set out in the patent had to be reformulated because it had **not been credibly solved**.

In **T 2341/13** the invention related to the hardware implementation of an interleaver. The examining division had considered it problematic that the application suggested that the invention could be used in a communication system based on a standard that was neither publicly available at the priority date nor fully disclosed in the application. The board observed that knowledge of any communication standard was not necessary to carry out the claimed invention and that it was perfectly valid to pose the problem of obtaining interleavers for frame sizes that were not a multiple of 2**(m). The board found that whether the application sufficiently disclosed the advantages of such frame sizes was irrelevant, unless it was argued that the mere idea of using such frame sizes was itself inventive (in which case it could not be included in the problem formulation).

In **T 1861/17** the board conceded that some decisions had taken the problem set out in the patent ("subjective problem") as their starting point for defining the objective technical problem (see e.g. **T 246/91**, **T 495/91**, **T 606/99**) but considered that this "objective problem" should normally be formulated only once the closest prior art had been identified. It was only on the basis of the features distinguishing the claimed invention from the closest prior art that the objective technical problem could actually be established in accordance with the established problem-solution approach (see e.g. **R 9/14**).

In **T 1099/16** the board stated that the extent to which a new technical effect underlying the claimed new purpose must be "described in the patent" for the exercise of construing the wording of a claim did not involve considerations of whether the technical effect was sufficiently credibly or plausibly described in the patent, but merely whether it had been described in the sense that a skilled person could recognise what technical effect was underlying the new purpose claimed. The decision whether a technical feature of a claim could be considered to be described in the patent therefore remained to be decided on the facts of the individual case.

4.2.3 Formulation of partial problems – lack of unity

In **T 314/99** the three different embodiments which were covered by claim 1 did not belong to the same single general inventive concept (Art. 82 EPC 1973). According to **G 1/91** (OJ 1992, 253) lack of unity is not an issue in opposition (or opposition appeal) proceedings. In the case in point the board stated that the consequence of this conceptual lack of unity was that different aspects of the problem applied to the three embodiments and that where conceptual non-unity arises between different embodiments covered by a claim, this may necessitate the formulation of corresponding partial problems, the respective solutions of which must be assessed separately for inventive step. With respect to the requirements of Art. 56 EPC 1973 the inventiveness of the subject-matter of a claim must be denied as a whole in the event that only one of its embodiments is obvious.

4.3. Solving a technical problem

The boards have regularly considered in the context of the evaluation of inventive step whether or not the problem is credibly solved. A great number of decisions address the assessment of alleged advantages, comparative tests and post-published documents.

4.3.1 Sufficient evidence for alleged advantages

According to the established case law of the boards of appeal, alleged advantages to which the patent proprietor/applicant merely refers, without offering sufficient evidence to support the comparison with the closest prior art, cannot be taken into consideration in determining the problem underlying the invention and therefore in assessing inventive step (see e.g. **T 20/81**, OJ 1982, 217; **T 181/82**, OJ 1984, 401; **T 1051/97**; **T 632/03**; **T 1211/07**; **T 736/12**; **T 2400/12**; **T 825/18**; **T 2210/19**). In **T 1027/08** the board added that there was no reason to deviate from this case law as it was based on the understandable rule that a patent could only properly be granted for a solution claimed as non-obvious if it actually achieved the alleged effect.

If the patent proprietor/applicant alleges the fact that the claimed invention improves a technical effect, then the burden of proof for that fact rests upon him (**T 355/97**, **T 1213/03**, **T 1097/09**, **T 2418/10**, **T 1487/16**). In the absence of any data confirming the alleged improvement, such an effect cannot be taken into account in the formulation of the technical problem (**T 2044/09**).

In **T 524/17** the board found it was not credible that the problem was solved over the whole range claimed. It held that where the closest prior art already showed improved values for certain properties which were said in the patent to be improved, it was up to the proprietor to show by means of suitable evidence that there was indeed such an improvement. In the absence of such evidence, it must be concluded that there was no such improvement and, thus, the problem was not solved. In **T 1346/16** the board concluded that the claimed subject-matter could be plausibly considered to have solved the technical problem ("technically plausible").

In **T 946/16** the board rejected the patent proprietor's formulation of the problem because it defined the quality of the copolymers obtained by the claimed method purely in absolute terms ("high quality"). To arrive at a an objective definition of the problem actually solved as compared with the closest prior art (D2), it instead had to be established whether the copolymer quality achieved using the claimed method was higher, similar or lower than that achieved in D2. In view of the technical arguments and the evidence adduced, the board considered that the patent proprietor had failed to show that the alleged improvement in copolymer quality over that achieved using the prior-art method was credible.

In **T 2514/16** the board found that the respondent had demonstrated with D16 that an effect was achieved for at least part of the claim at issue and that the burden of proving that this effect was not achievable across the whole breadth of the claim thus lay with appellant 1. Appellant 1 contended that, according to **T 1188/00**, it was for the patent proprietor to prove this. The board, however, considered that the reason that conclusion had been reached in **T 1188/00** was that the effect (improvement) alleged by the patent proprietor there had been found to be not credible, and so the circumstances were not comparable with those now at issue. In the absence of relevant comparative tests, appellant 1's objection was held to be unconvincing.

In T 575/17 the board held that, where there was no direct or plausible disclosure in the application as a whole of how the envisaged effect of the invention was achieved or of why the problem was solved by the claimed features, how this worked could also be deduced from the teaching of other (pre-published) documents. In the case in hand, the board, referring to documents E1 to E3 and E5, observed that using prefabricated implants for better thermal conductivity had long been known in detail. The skilled person would have no technical difficulty in applying the teaching of these documents to the "insulated metal substrate" technology that had later become established. It did not matter that the documents recorded an "old technology"; they contained the general technical teaching that an object inserted into the metal (substrate) could be used to improve the conduction of an electrical component's heat.

4.3.2 Comparative tests

As early as T 35/85 it was stated by the the board that an applicant or patentee may discharge his onus of proof by voluntarily submitting comparative tests with newly prepared variants of the closest state of the art identifying the features common with the invention, in order to have a variant lying closer to the invention so that the advantageous effect attributable to the distinguishing feature is thereby more clearly demonstrated (T 40/89, T 191/97, T 496/02, T 765/15, T 1323/17).

According to the established case law, an unexpected effect (advantageous effect or feature) demonstrated in a comparative test can be taken as an indication of inventive step (T 181/82, OJ 1984, 401). In T 197/86 (OJ 1989, 371) the board supplemented the principles laid down in T 181/82, according to which, where comparative tests were submitted as evidence of an unexpected effect, there had to be the closest possible structural approximation in a comparable type of use to the subject-matter claimed. It stated that if comparative tests are chosen to demonstrate an inventive step on the basis of an improved effect over a claimed area, the nature of the comparison with the closest state of the art must be such that the alleged advantage or effect is convincingly shown to have its origin in the distinguishing feature of the invention compared with the closest state of the art (see also T 234/03, T 568/11, T 1457/13, T 1521/13; T 1401/14; T 710/16, T 990/17, T 2406/18, T 816/16)

For this purpose it might be necessary to modify the elements of comparison so that they differed only by such a distinguishing feature (see e.g. T 197/86, T 292/92, T 819/96, T 369/02, T 2043/09, T 183/12, T 710/16, T 990/17).

To be of relevance in demonstrating that a technical improvement is achieved in comparison with the closest state of the art, any comparative test presented must be reproducible on the basis of the information thus provided, thereby rendering the results of such tests directly verifiable (T 494/99, T 234/03, T 236/09, T 1962/12, T 383/13). This requirement implies, in particular, that the procedure for performing the test relies on quantitative information enabling the person skilled in the art to reproduce it reliably and validly (T 234/03, T 236/09, T 383/13, T 1962/12, T 532/14, T 795/14). Vague and imprecise operating instructions render the test inappropriate and thus irrelevant (T 234/03, T 236/09, T 383/13, T 1962/12, T 795/14).